Premenstrual Syndrome (PMS)

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Abstract. Premenstrual Syndrome (PMS) is not just hormonal; heavy metals and gut bacteria play crucial roles. Managing these factors with diet, probiotics, and toxin avoidance can improve your symptoms.

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Overview

Premenstrual Syndrome (PMS) is a common condition affecting women of reproductive age, characterized by a range of physical, psychological, and behavioral symptoms that occur in the luteal phase of the menstrual cycle. Symptoms typically subside after menstruation begins. It affects approximately 47.8% of women globally, with the most severe form, Premenstrual Dysphoric Disorder (PMDD), impacting 3-8% of women.[1] Common symptoms include mood swings, irritability, bloating, fatigue, and breast tenderness.[2] The exact etiology of PMS remains unclear, but hormonal fluctuations, particularly involving progesterone and estrogen, as well as neurotransmitter imbalances (including serotonin and GABA), play a central role in its development.[3] The microbiome's role in PMS has not been extensively explored, but some studies suggest that gut microbiota imbalances may influence hormonal regulation and mood disorders associated with PMS, suggesting a potential avenue for future research.[4][5]

Causes

The exact cause of PMS remains unclear, but several theories attempt to explain the underlying mechanisms. The most widely accepted theory involves hormonal fluctuations, particularly changes in progesterone and estrogen levels during the luteal phase.[6] These hormonal changes are thought to influence neurotransmitter systems, including serotonin and GABA, which regulate mood and behavior.[7] More recent studies have highlighted the role of the neurosteroid allopregnanolone, a metabolite of progesterone, which modulates GABA-A receptors and may contribute to the mood symptoms of PMS.[8] Research also suggests that women with PMS may have an abnormal response to normal hormonal fluctuations, possibly due to genetic predispositions or sensitivity to changes in hormone levels.

Diagnosis

Diagnosing PMS involves the identification of cyclic symptoms, which should appear in the luteal phase and resolve with menstruation.[9] It is important to rule out other conditions that could mimic PMS, such as thyroid disorders or other psychiatric conditions. Microbiome signatures have shown promise in diagnosing PMS, with studies indicating that imbalances in gut microbiota may influence hormonal regulation and the severity of PMS symptoms.[10] Other potential diagnostic methods include metabolomic

and metallomic signatures, which could offer non-invasive diagnostic options for identifying PMS by tracking specific biomarkers associated with hormonal changes and neurotransmitter imbalances.

Associated Conditions

PMS is closely associated with Premenstrual Dysphoric Disorder (PMDD), a more severe form of PMS that significantly disrupts daily functioning.[11] PMDD shares similar symptoms with PMS but includes more intense psychological manifestations, such as severe depression, irritability, and anxiety.[12] Both conditions have been linked to hormonal fluctuations and neurotransmitter imbalances, particularly involving serotonin and GABA. Additionally, PMS and PMDD are often associated with other mood disorders, such as anxiety and depression, which may complicate diagnosis and treatment.[13] These associations highlight the complex, multifactorial nature of PMS, involving not only hormonal changes but also psychological and social factors.

Primer

Premenstrual Syndrome (PMS) is a complex condition affecting women of reproductive age, characterized by a combination of physical, psychological, and behavioral symptoms that occur in the luteal phase of the menstrual cycle.[14] These symptoms typically subside with the onset of menstruation. The pathophysiology of PMS is not fully understood, but hormonal fluctuations are central to the development of symptoms.[15] Recent studies also highlight the potential role of the gut microbiota in influencing the severity of PMS.[16] Alterations in the gut microbiome may affect hormone regulation, neurotransmitter function, and the body's inflammatory response, potentially contributing to the mood swings, irritability, and physical discomfort observed in PMS.[17] Research suggests that metallomic factors, such as exposure to heavy metals like copper, magnesium and zinc, may also play a role in regulating hormones and influencing PMS symptoms through oxidative stress pathways.[18] Given these multifactorial influences, understanding the intersections between hormonal changes, the gut microbiome, and metallomic signature is crucial for developing a more comprehensive and personalized approach to treating PMS.

Metallomic Signature

Studies suggest that these metals influence hormonal regulation, immune function, and neurochemical pathways, which are disrupted in PMS, thereby exacerbating symptoms such as mood swings, irritability, fatigue, and physical discomfort. These metal imbalances are further complicated by environmental and lifestyle factors, notably smoking, which significantly increases the burden of these metals in the body.[19][20] The presence and concentrations of these metals can directly affect neurotransmitter function and inflammation, key mechanisms in the pathophysiology of PMS. Both essential and toxic metals can influence various biological systems that regulate mood, immune function, and hormone production. For instance, zinc (Zn) and iron (Fe) are essential for metabolic processes but can become harmful when imbalanced, while metals like cadmium (Cd) and lead (Pb) are toxic and have been shown to disrupt hormonal balance and contribute to PMS severity.[21] Tobacco consumption is a significant source of heavy metals, including cadmium, lead, manganese, and iron, all of which contribute to the metallomic signature of PMS.[22][23] The heavy metals then accumulate in the body and can exacerbate the hormonal and neurochemical imbalances that define PMS. Other sources of exposure to these metals include environmental pollution, occupational hazards, contaminated food, water, and even household products.

Interventions

Intervention	MBTI Status	Mechanism of Action
Probiotics	Validated	Probiotics, which help balance the gut microbiota, are emerging as a potential option for managing premenstrual disorders (PMDs) by impacting microbial diversity, metabolic activity, and inflammation.[24] Research suggests that specific probiotic strains can modulate cytokine levels, support immune tolerance, and ease symptoms linked to hormone-related conditions. Since conventional treatments can cause side effects and may not always work, probiotics offer a safer and possibly more effective alternative.[25][26]
Prebiotics	Promising Candidate	Prebiotics, which include dietary fibers serve as food for beneficial gut bacteria. By promoting the growth of these bacteria, prebiotics improve the balance of the gut microbiome, leading to enhanced production of SCFAs like acetate, propionate, and butyrate. These SCFAs help regulate immune function, reduce gut inflammation, and influence hormonal signaling pathways, which may contribute to alleviating PMS symptoms such as bloating, irritability, and fatigue.[28][x][29][x]
Fecal Microbiota Transplantation (FMT)	Experimental	FMT aims to restore a healthy and diverse microbiome, which can reduce systemic inflammation, regulate immune responses, and potentially alleviate hormonal imbalances seen in PMS. By improving gut flora, FMT can enhance the gut-brain axis and mitigate symptoms such as mood swings and anxiety.[30][x]
Synbiotics	Promising Candidate	Synbiotics combine both probiotics and prebiotics to maximize gut microbiome modulation. The probiotics enhance the growth of beneficial bacteria, while the prebiotics provide nutrients that support their survival and activity. This combined effect helps restore microbiome balance, promote SCFA production, and improve gut permeability. Synbiotics may also enhance serotonin production and reduce inflammation, thus helping to alleviate both the physical and emotional symptoms of PMS.[31][x][32][x]
Dietary Interventions	Promising Candidate	Diets high in calcium, magnesium, vitamin D, B vitamins, zinc, and omega-3 fatty acids reduce PMS symptom severity by modulating neurotransmitter synthesis (e.g., serotonin), influencing hormone metabolism, and reducing oxidative stress and inflammation. Complex carbohydrates and tryptophan enhance serotonin availability. Western diets rich in sugar, salt, trans fats, and processed foods worsen symptoms. High-fiber, plant-based diets and regular, small meals stabilize blood glucose, improve mood, and reduce bloating and irritability.[33][x]
Short-Chain Fatty Acids (SCFAs) Supplementation	Under Investigation	SCFAs like acetate, propionate, and butyrate are produced by gut bacteria when they ferment dietary fibers. These SCFAs serve as an energy source for colonocytes and play a crucial role in regulating immune function and reducing inflammation. SCFAs also enhance

Intervention	MBTI Status	Mechanism of Action
		the gut barrier function, preventing leaky gut, and modulating gut-brain communication. SCFA supplementation could help balance the microbiome, regulate hormonal levels, and reduce PMS symptoms such as mood disturbances and cramps. [34][35]
Excercise	Promising Candidate	Exercise helps alleviate PMS by temporarily increasing estrogen and progesterone, reducing hormonal withdrawal symptoms. It decreases fluid retention through lowered renin and aldosterone, relieves pain via prostaglandin production, and enhances mood and reduces stress through endorphin release.[36][37]

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